

14.1.1. Complex Number Addition using Class

Aim: Write a Python program using a class named **Complex** to add two complex numbers.

Algorithm

Step 1: Start

Step 2: Create three objects: c1, c2, c3 of class Complex

Step 3: Input first complex number: Read real and imaginary part
Store in c1.real and c1.imag

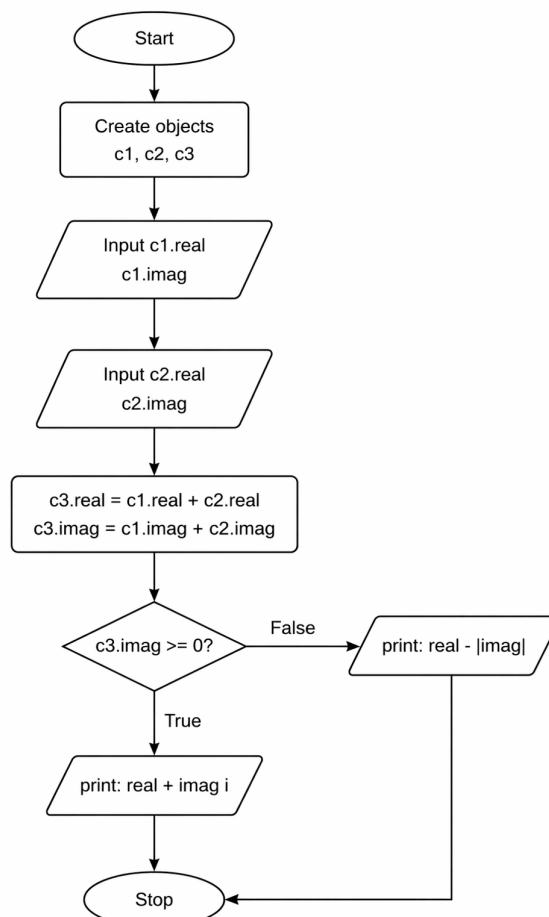
Step 4: Input second complex number: Read real and imaginary
part Store in c2.real and c2.imag

Step 5: Compute sum: $c3.real \leftarrow c1.real + c2.real$ $c3.imag \leftarrow c1.imag + c2.imag$

Step 6: Display result: If $c3.imag \geq 0$ Print: $c3.real + c3.imag \ i$ Else
Print: $c3.real - |c3.imag| \ i$

Step 7: Stop

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14.1.1. Complex Number Addition using Class

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Write a Python program using a class named **Complex** to add two complex numbers.

Create a class **Complex** with the following methods:

- initComplex()**: Read user input for the real and imaginary parts.
- sum()**: Compute the sum of two complex numbers and store the result in the current instance.
- display()**: Display the complex number in the format "<a> + i" or "<a> - i".

The program should create three instances of **Complex** class: two for input complex numbers and one for the result. Initialize both complex numbers, compute their sum, and display the result.

Input Format:

- First line contains two space-separated numbers: real and imaginary parts of first complex number
- Second line contains two space-separated numbers: real and imaginary parts of second complex number

Output Format:

- Print a single line showing the sum of the two complex numbers in the format "<a> + i" or "<a> - i".

Sample Test Cases

complex...

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1class Complex:
2 def initComplex(self):
3 self.real, self.imag = map(int, input().split())
4 def sum(self, c1, c2):
5 self.real = c1.real + c2.real
6 self.imag = c1.imag + c2.imag
7 def display(self):
8 if self.imag >= 0:
9 print(f"{self.real} + {self.imag}i")
10 else:
11 print(f"{self.real} - {abs(self.imag)}i")
12
13
14# Create three instances
15c1 = Complex()
16c2 = Complex()
17c3 = Complex()
18
19# Initialize two complex numbers
20c1.initComplex()
21c2.initComplex()
22

Average time0.003 s3.00 msMaximum time0.009 s9.00 ms

4 out of 4 shown test case(s) passed
4 out of 4 hidden test case(s) passed

Test case 19 msDebug
Expected output
3 4
2 5
5 + 9i
Actual output
3 4
2 5
5 + 9i
Test case 23 ms

TerminalTest cases

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